Math 114 Class notes
Fall 2026

To accompany
Precalculus 2e
by *Abramson*

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Math 114 Formula Sheet

Unit One

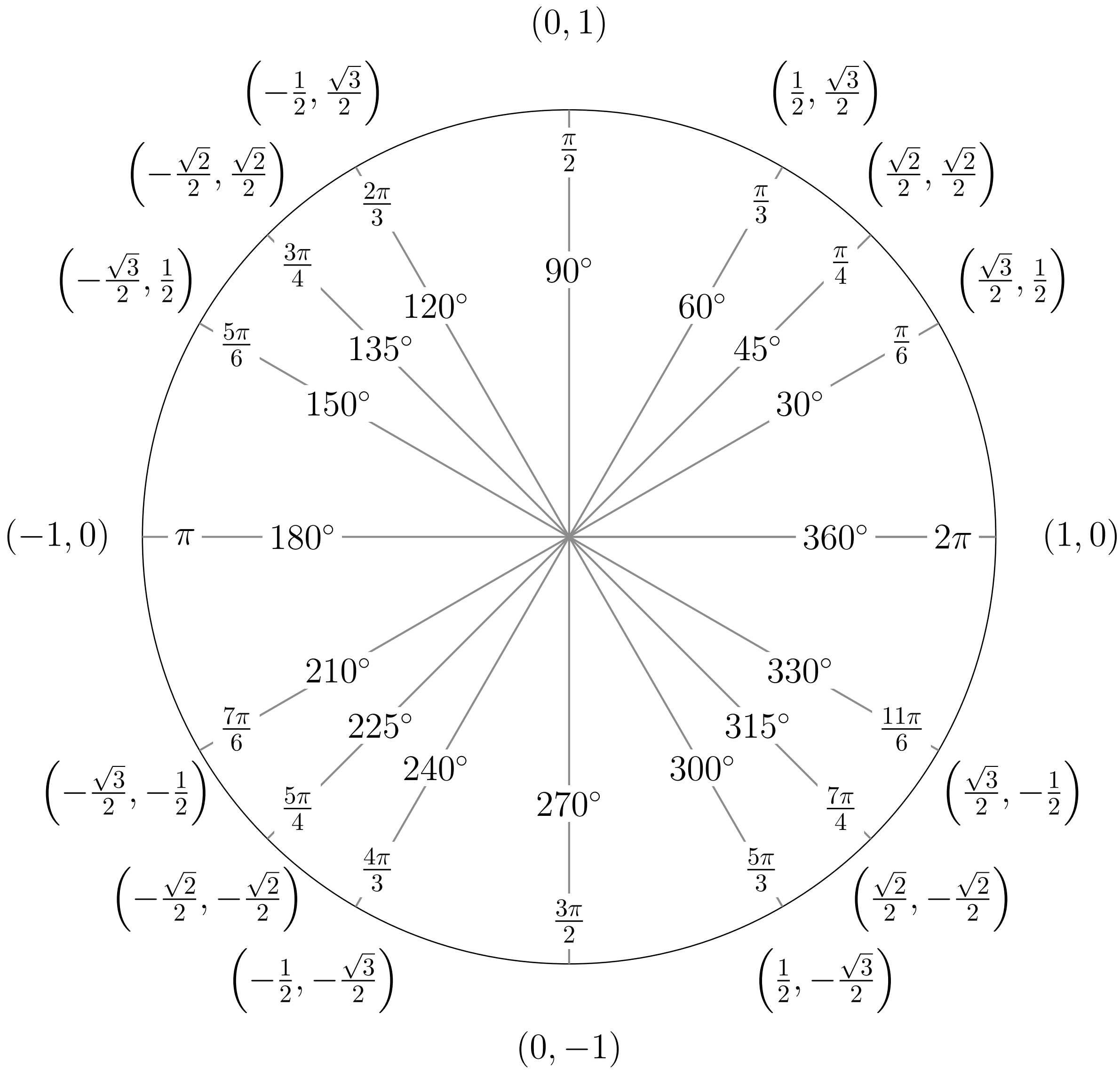
Fractional exponents: $x^{a/b} = \sqrt[b]{x^a}$

Subtracting exponents: $x^{a-b} = \dfrac{x^a}{x^b}$

Average Rate of Change: $\dfrac{f(b) - f(a)}{b - a}$

Unit Circle

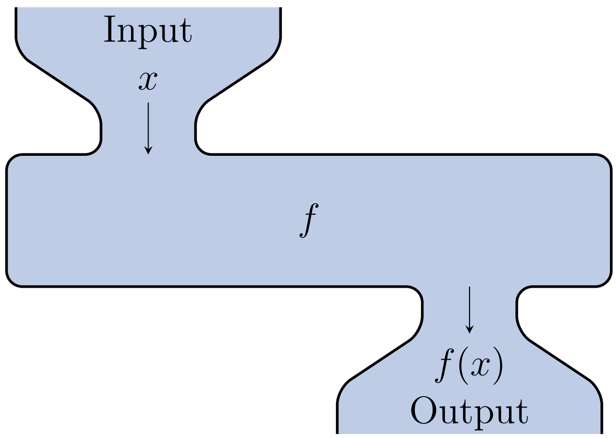
$$\big(\cos(\theta),\ \sin(\theta)\big)$$

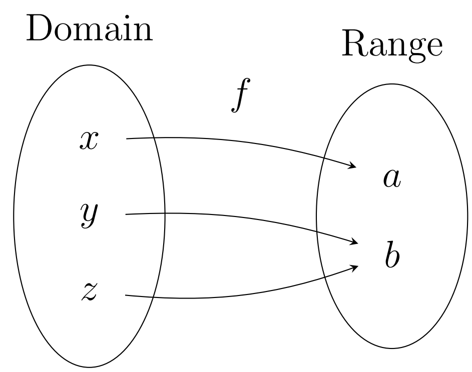


$$180^\circ = \pi \text{ rad}$$

1.1: Functions and Function Notation

Definition.
A **function** is a relation in which each possible input value leads to exactly one output value.
The input values make up the **domain**, and the output values make up the **range**.





Example. Let $f(x) = 2x^2 - 2x + 1$. Evaluate the following

$f(1)$ $\qquad$ $f(-2)$ $\qquad$ $f(a)$

$f(a+h)$ $\qquad$ $\dfrac{f(a+h) - f(a)}{h}$

Example. Suppose that the following tables represent a carnival truck's menu prices. Which table would represent a valid function?

Item	Price
Funnel cake	\$8
Chicken strips	\$9
Ribbon fries	\$7
Cheese nachos	\$6

Item	Price
Funnel cake	\$8
Chicken strips	\$8
Ribbon fries	\$7
Cheese nachos	\$6

Item	Price
Funnel cake	\$8
Chicken strips	\$8
Ribbon fries	\$8
Funnel cake	\$6

Example. Based on an old match-makers rule, the socially "acceptable" minimum age of someone you can date is half your age plus seven. Make a function that takes your age, x, and finds the minimum age of someone you can date, y.

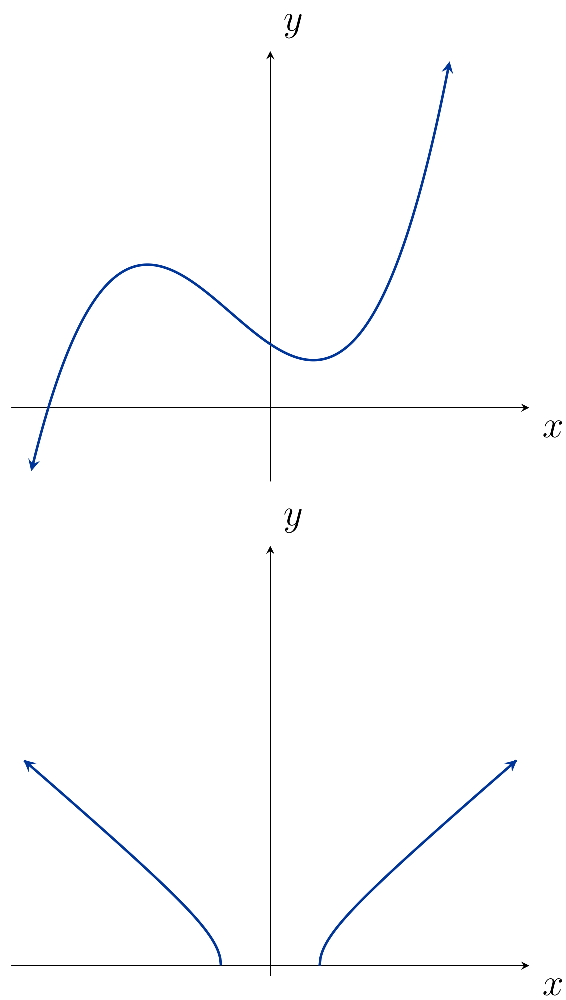
Example. Create a table to represent a function that takes each month and returns the number of days (in a non-leap year).

Example. If possible, express the relationship $2n + 6p = 12$ as a function $p = f(n)$. Graph the result.

Example. If possible, express the relationship $x^2 + y^2 = 1$ as a function $y = f(x)$. Graph the result.

Definition. (Vertical Line Test)
A curve in the xy-plane is the graph of a function $y = f(x)$ (an explicit function) if and only if each vertical line intersects it in at most one point

Example. Use the vertical line test on the following graphs to determine which graphs may represent an explicit function:



Definition. (One-to-One)
The **horizontal line test** determines if every horizontal line passes through the graph of a function in at most one point.
A function is said to be **one-to-one** if each output value corresponds to *exactly* one input value (and hence passes the horizontal line test).

Example. Which of the following represent a one-to-one function? Graph

$y = x^2$

$y^2 = x$

$f(x) = \sqrt{x}$

$y = |x|$

$f(x) = (x+1)(x-1)(x-2)$

The equation of a circle:
$x^2 + y^2 = 1$

The area of a circle:
$A = \pi r^2$

Graphs of basic functions

1.2: Domain and Range

Definition. (Interval notation)
We can specify the domain and/or range of a function using interval notation:

- Strict inequalities (exclusive) are represented using parenthesis:

$$x > 3 \iff (3, \infty) \qquad\qquad x < 9 \iff (-\infty, 9)$$

  Note: Parenthesis are **always** used when an interval's end points are infinite

- Nonstrict inequalities (inclusive) are represented using square brackets:

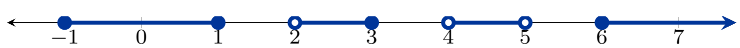
$$7 \leq x \leq 11 \iff [7, 11]$$

- Disjoint intervals can be unioned together:

$$x \neq -2 \iff (-\infty, -2) \cup (-2, \infty)$$

- Any combination of the above can be used:

$$[-1, 1] \cup (2, 3] \cup (4, 5) \cup [6, \infty)$$



Recall that the domain is the set of all *possible* input values.

Example. Find the domain of the following functions (use interval notation): Graphs

$f(x) = x - 3$

$A = \pi r^2$

$y = \sqrt{x-1}$

$y = \dfrac{x+2}{x^2-4}$

$y = \sqrt[3]{x-4}$

$f(x) = \dfrac{\sqrt[4]{x^2+4}}{(x+1)\sqrt{x-1}}$

Example. Identify the domain and range of the following: Graphs

$x(x+1)(x+2)^2, \quad x \geq -2$

$\sin(x)$

$\log(x)$

$2^x, \quad x \geq -1$

Definition.
A **piecewise** function is a function with different definitions for different portions of the domain.

Example. Rewrite the following as piecewise functions:

$|x| =$

$\dfrac{x}{|x|} =$

$|x-1| + |4-x| =$

Example. Graph the following piecewise function: Graphs

$$f(x) = \begin{cases} x^2, & \text{if } x \leq 1 \\ 3, & \text{if } 1 < x \leq 2 \\ x, & \text{if } x > 2 \end{cases}$$

1.3: Rates of Change and Behavior of Graphs

Definition. (Rate of Change)
A rate of change describes how an output quantity changes relative to the change in the input quantity. The units on a rate of change are "output units per input units". The **average rate of change** between two input values is the total change of the function values divided by the input values.

$$\frac{\Delta y}{\Delta x} = \frac{f(x_2) - f(x_1)}{x_2 - x_1}$$

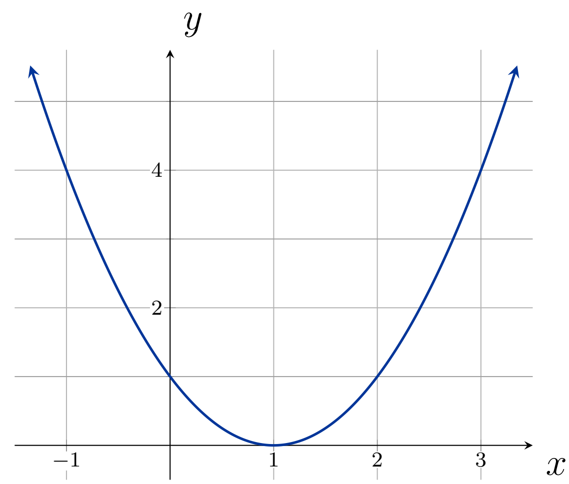
Example. Suppose the following table contains temperatures recorded through the day.

h	7 AM	9 AM	11 AM	1 PM	3 PM
$T(h)$	$78°F$	$81°F$	$87°F$	$93°F$	$98°F$

Find the average rate of change between

7 AM and 9 AM   9 AM and 11 AM   7 AM and 3 PM

Example. Using the following graph, determine the average rate of change on the interval $[-1, 2]$



Example. Find the average rate of change of $f(x) = x^2 - \dfrac{1}{x}$ on the interval $[2, 4]$.

Example. Find the expression for the average rate of change of $g(t) = t^2 + 3t + 1$ on the interval $[0, a]$.

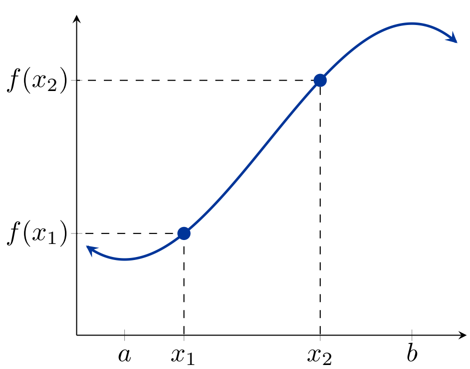
Example. Find the expression for the average rate of change of $j(t) = \dfrac{1}{t+3}$ on the interval $[1, 1+h]$.

Example. Find the expression for the average rate of change of $g(x) = \sqrt{x-1}$ on the interval $[x, x+h]$.

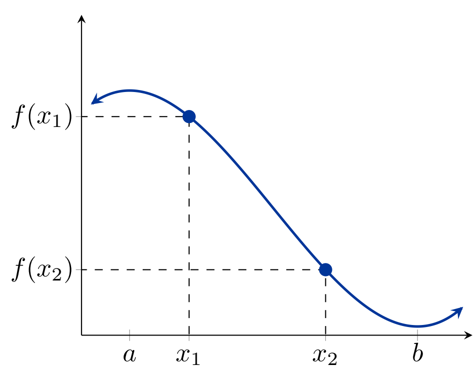
Definition.

Consider the function $f(x)$ on the interval (a, b). Given *any* two numbers x_1 and x_2 in (a, b) where $x_1 < x_2$, we say f is

increasing if $f(x_1) < f(x_2)$

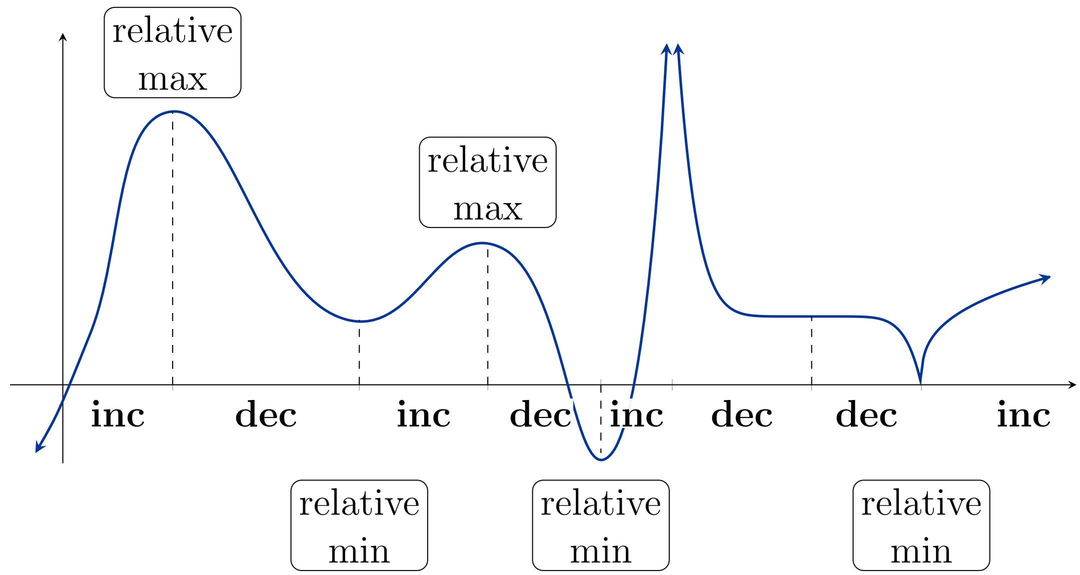


decreasing if $f(x_1) > f(x_2)$



Definition. (Relative/Local Extrema)
A function f has a

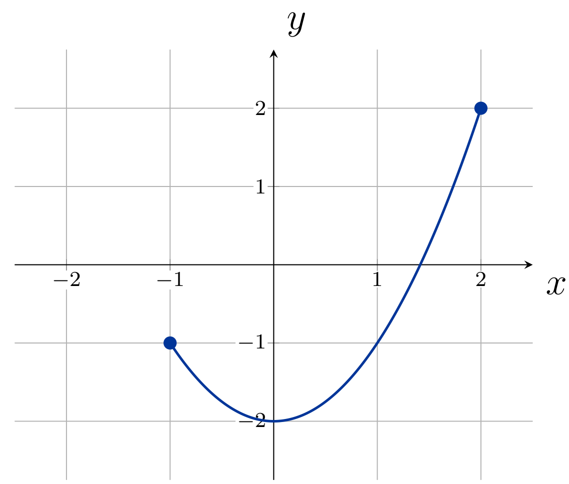
- **relative maximum** at $x = c$ if $f(c) \geq f(x)$ for every x in (a, b)
- **relative minimum** at $x = c$ if $f(c) \leq f(x)$ for every x in (a, b)



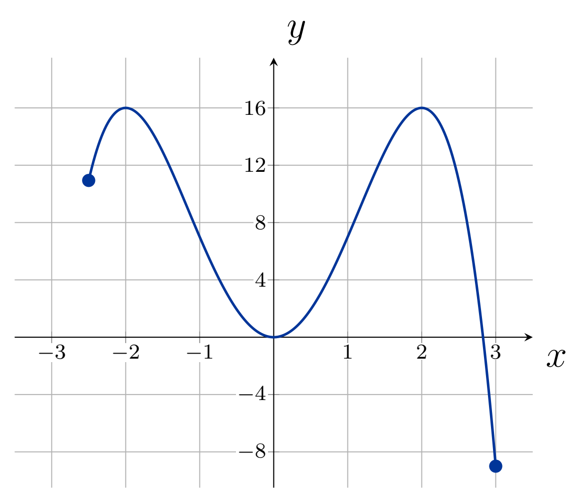
Definition. (Absolute Extrema)
A function f defined on a domain D has an

- **absolute maximum** at $x = c$ if $f(c) \geq f(x)$ for every x in D
- **absolute minimum** at $x = c$ if $f(c) \leq f(x)$ for every x in D

Example. Using the following graph, find any relative and absolute extrema.



Example. Using the following graph, find any relative and absolute extrema.



1.4: Composition of Functions

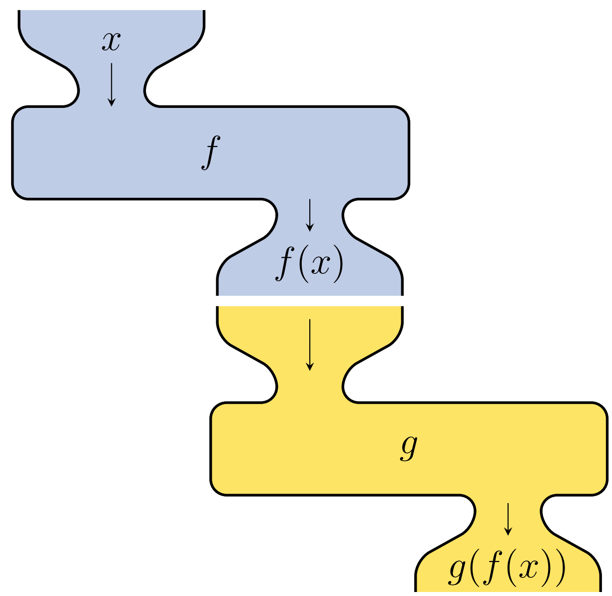
Definition. (The Composition of Two Functions)
Let f and g be functions. Then the composition of g and f is the function $g \circ f$ defined by

$$(g \circ f)(x) = g(f(x))$$

The domain of $g \circ f$ is the set of all x is the domain of f such that $f(x)$ lies in the domain of g.

Example. Let $f(x) = \sqrt{x+1}$ and $g(x) = 4 - x$. Find the domain of the following:

$g(f(x)) =$



$f(g(x)) =$

$f(f(x)) =$

Example. For the following, find functions $g(x)$ and $h(x)$ so that $f(x) = g(h(x))$:

$$f(x) = \frac{1}{x+2} \qquad\qquad f(x) = \sqrt{x-4}$$

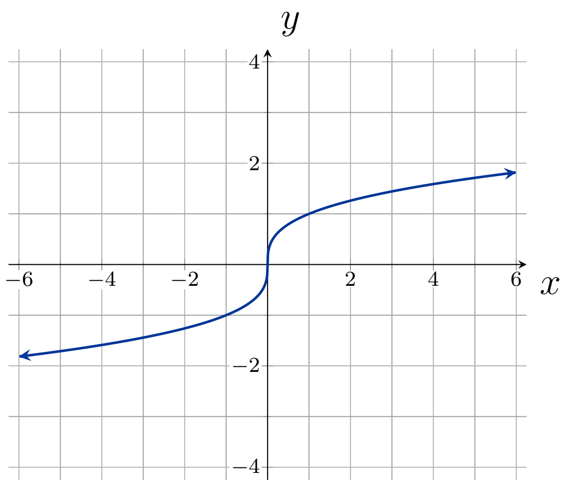
$$f(x) = \left(x^2 + 3x - 8\right)^4 \qquad\qquad f(x) = \frac{x-3}{\sqrt{-(x-3)}}$$

1.5: Transformation of Functions

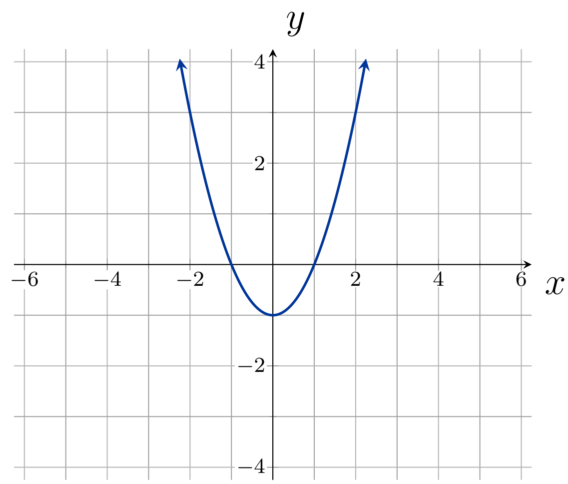
Definition. (Vertical Shift)
Given a function $f(x)$, a new function $g(x) = f(x) + k$, where k is a constant, is a **vertical shift** of the function $f(x)$. All the output values shift vertically by k units (**up** if $h > 0$, **down** if $h < 0$).

Example. Using the graphs below, sketch the shifted functions as indicated:

Graph $f(x) + 2$:



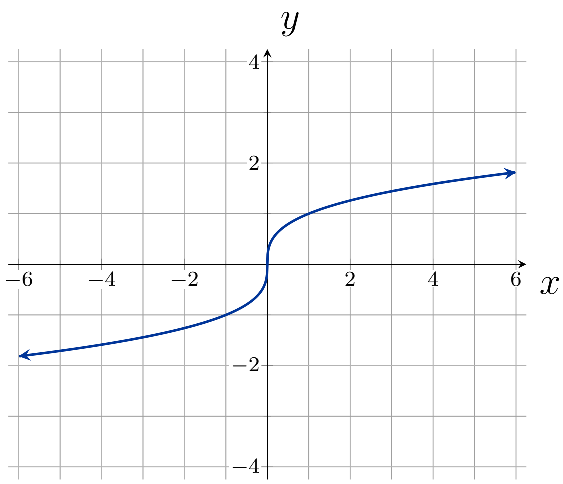
Graph $f(x) - 1$:



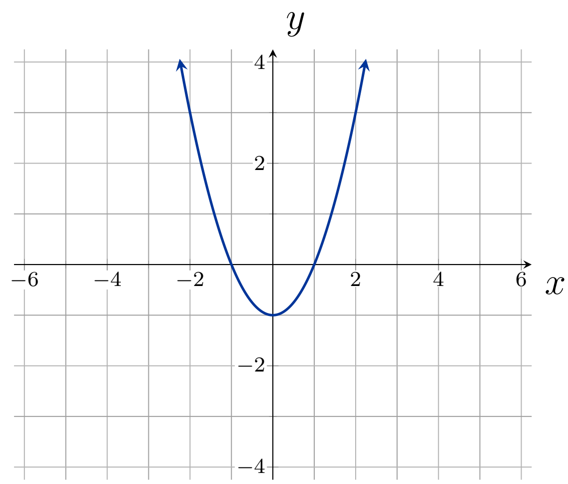
Definition. (Horizontal Shift)
Given a function $f(x)$, a new function $g(x) = f(x - h)$, where h is a constant, is a **horizontal shift** of the function $f(x)$. All the output values shift horizontally by h units (**right** if $k > 0$, **left** if $k < 0$).

Example. Using the graphs below, sketch the shifted functions as indicated:

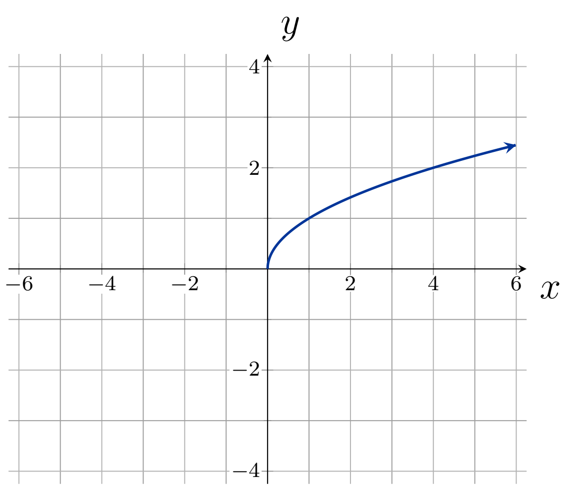
Graph $f(x+2)$:



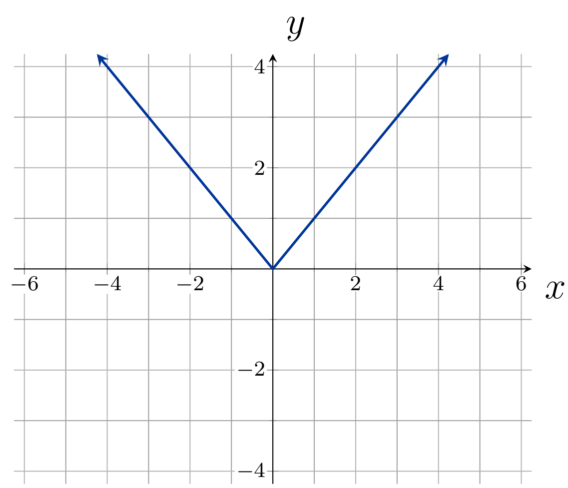
Graph $f(x-1)$:



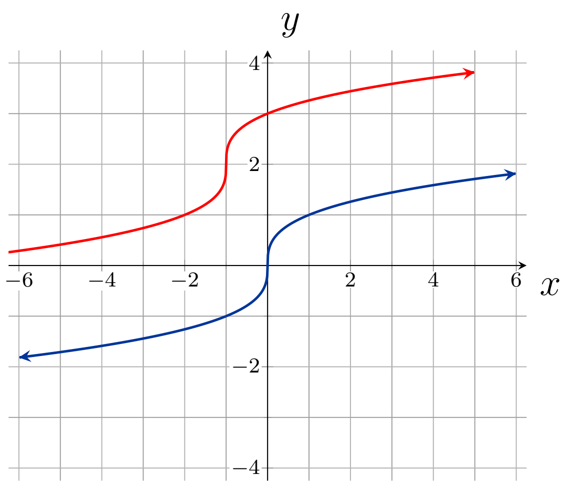
Graph $f(x-2)+1$:

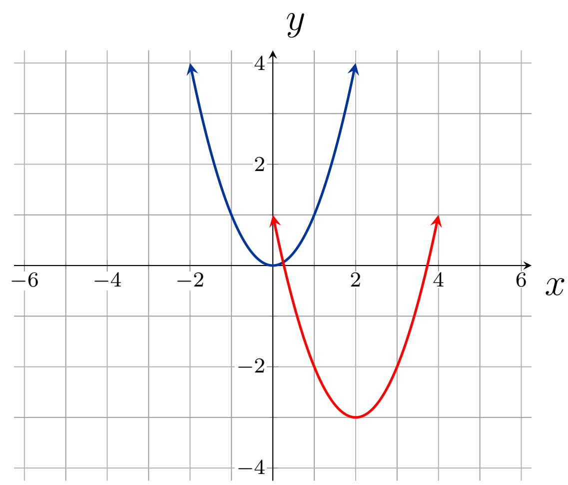


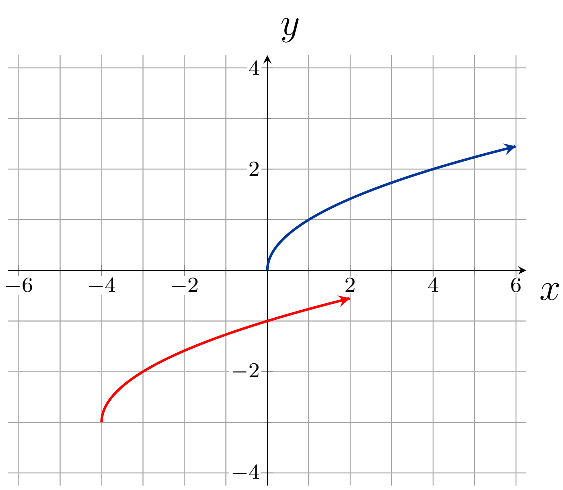
Graph $f(x+1)-3$:

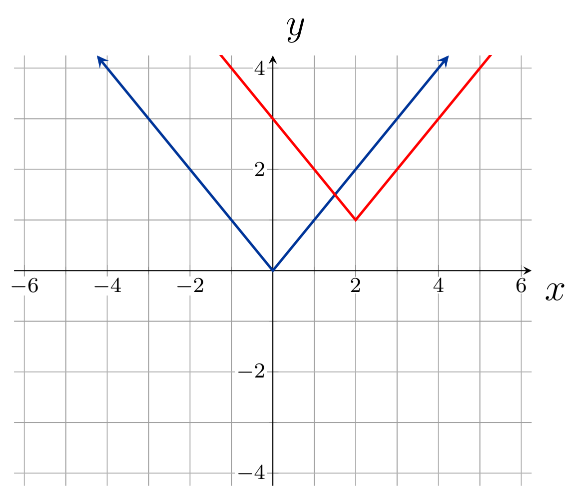


Example. Identify the shifts which transform the blue graph into the red graph:







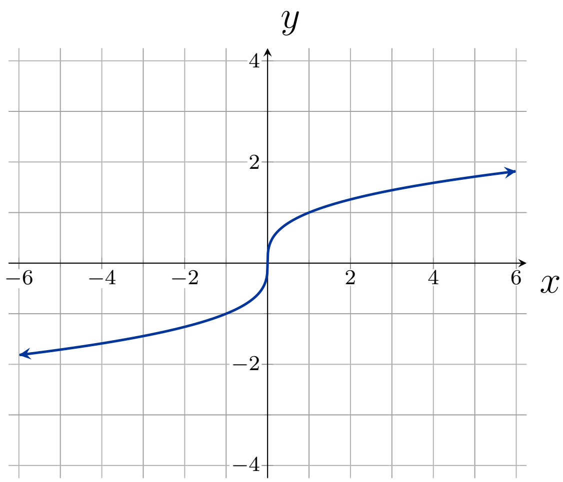


Definition. (Reflections)
Given a function $f(x)$, a new function $g(x) = -f(x)$ is a **vertical reflection** of the function $f(x)$, sometimes called a reflection about the x-axis.

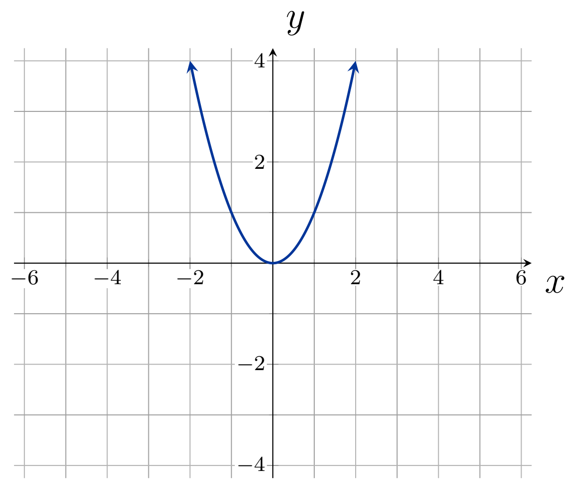
Given a function $f(x)$, a new function $g(x) = f(-x)$ is a **horizontal reflection** of the function $f(x)$, sometimes called a reflection about the y-axis.

Example. Transform the graphs below as indicated:

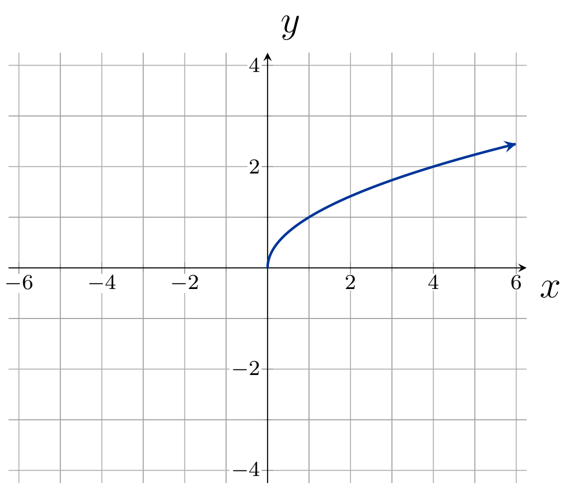
Graph $f(-x)$:



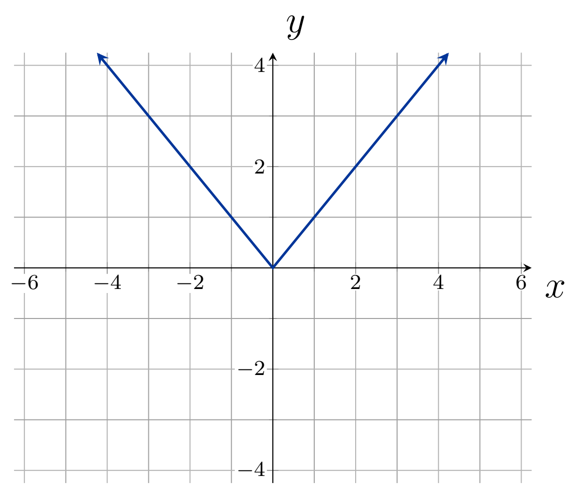
Graph $-f(x)$:



Graph $-f(-x)$:



Graph $-f(x) + 1$



Definition. (Even and Odd functions)
A function is called an even function if for every input x

$$f(x) = f(-x)$$

The graph of an even function is symmetric about the $y-$axis.

A function is called an odd function if for every input x

$$f(x) = -f(-x)$$

The graph of an odd function is symmetric about the origin.

Example. Using algebra, determine if the following functions are even, odd, or neither: Graphs

$f(x) = x^3$

$g(x) = |x| + 1$

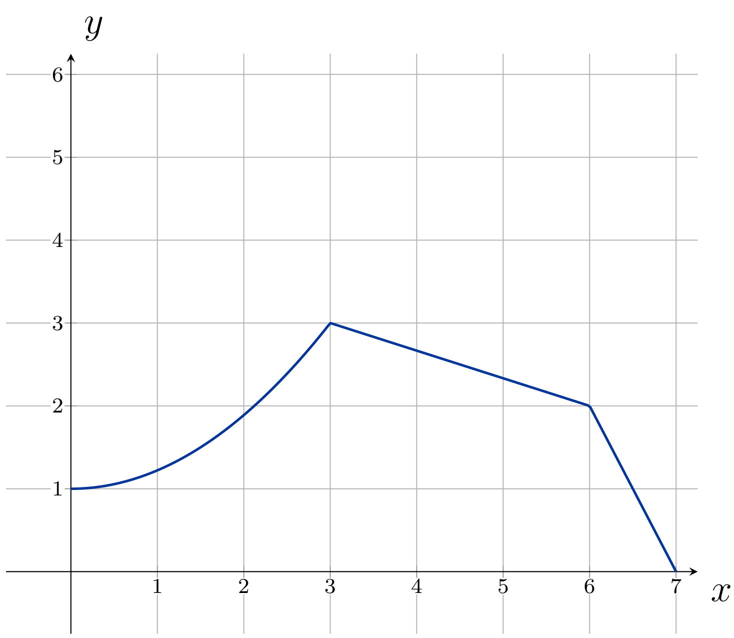
$h(x) = (x-1)^2$

$j(x) = x^3 - \dfrac{1}{x}$

Definition. (Vertical Stretches and Compressions)
Given a function $f(x)$, a new function $g(x) = af(x)$, where a is a constant, is a **vertical stretch** or **vertical compression** of the function $f(x)$.

- If $a > 1$, then the graph will be stretched.
- If $0 < a < 1$, then the graph will be compressed.
- If $a < 0$, then there will be a combination of a vertical stretch/compression with a vertical reflection.

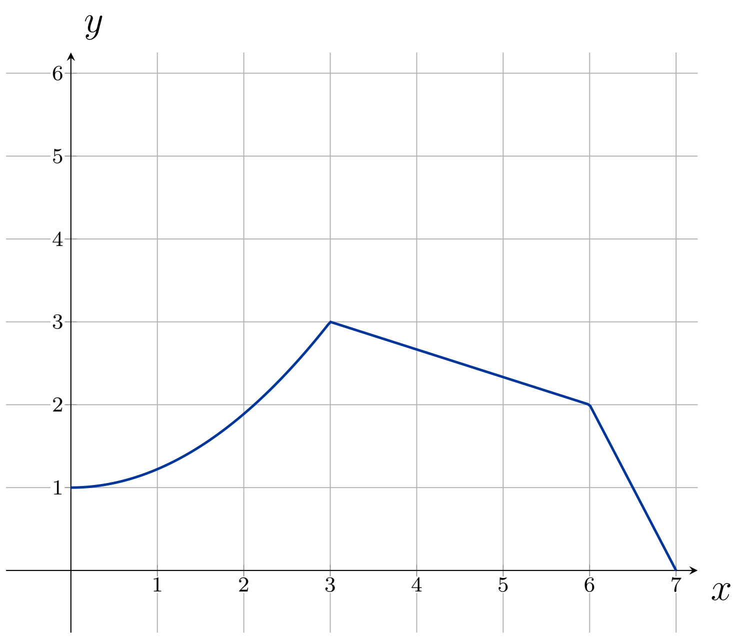
Example. Given the graph of $f(x)$ below, graph $g(x) = 2f(x)$: Graphs



Definition. (Horizontal Stretches and Compressions)
Given a function $f(x)$, a new function $g(x) = f(bx)$, where b is a constant, is a **horizontal stretch** or **horizontal compression** of the function $f(x)$.

- If $b > 1$, then the graph will be stretched by $\dfrac{1}{b}$.
- If $0 < b < 1$, then the graph will be compressed by $\dfrac{1}{b}$.
- If $b < 0$, then there will be a combination of a horizontal stretch/compression with a horizontal reflection.

Example. Given the graph of $f(x)$ below, graph $g(x) = f(2x)$: Graphs

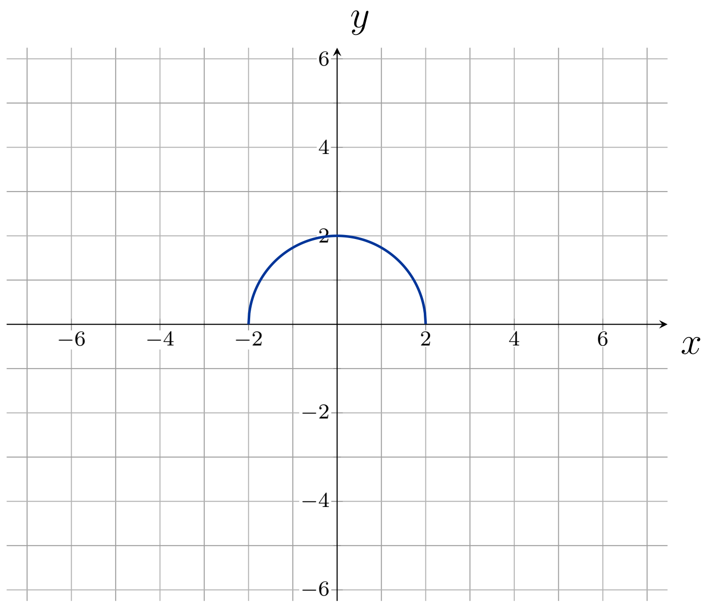


Combining Transformations

When combining transformations ...

- $af(x) + k$: first vertically stretch by a and then shift by k.
- $f(bx - h)$: first horizontally shift by h and then horizontally stretch by $\dfrac{1}{b}$.
- $f(b(x - h))$: first horizontally stretch by $\dfrac{1}{b}$ and then horizontally shift by h.
- Horizontal and vertical transformations are independent of the order they are performed.

Example. Given the graph of $f(x)$ below, graph $g(x) = f\left(\dfrac{1}{2}x + 1\right) - 3$: Graphs



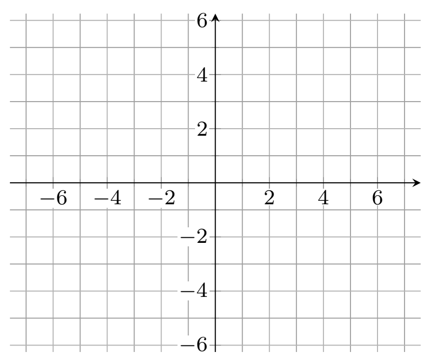
1.6: Absolute Value Functions

Definition. (Absolute Value Function)
The absolute value function can be defined as a piecewise function

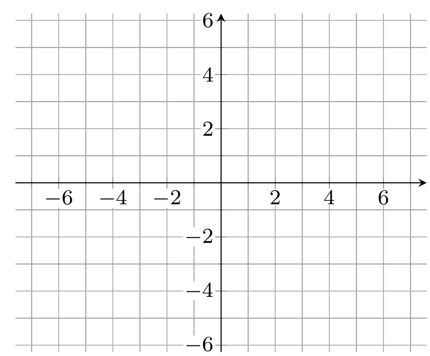
$$f(x) = |x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}$$

Example. Graph the following

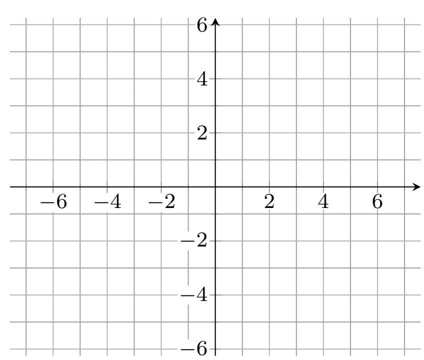
$f(x) = |x|$



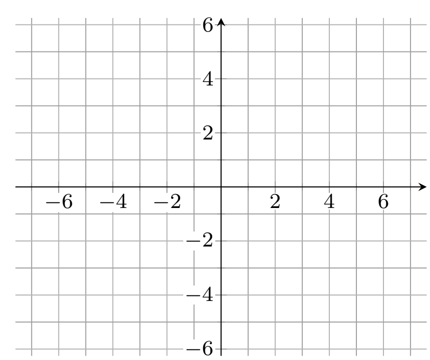
$f(x) = |x - 2|$



$f(x) = |x| + 3$



$f(x) = 2|x - 3| + 1$



Example. Solve the following equations Graphs

$|x| = 3$

$|2x + 6| = 2$

$|x + 3| - 1 = 0$

$2 - |4 - x| = 0$

Example. Solve the following inequalities. Give your answer in interval notation. Graphs

$|x - 5| \leq 4$

$\left|\frac{x}{4} - 1\right| < 2$

$|2 - x| \leq 3$

$|x - 3| > 2$